

UrbanPulse Task B Report

Task B Report

Problem Statement

Demonstrate Kafka cluster configuration, topic design, producer durability, validation, DLQ handling, consumer lag, and a stream-table join.

Design

Three brokers and replication factor 3 provide the needed durability; topics are sized by workload; producers use idempotence and delivery retries; validation errors route to the DLQ.

Implementation

See ``create_topics.py``, ``producers/aq_producer.py``, ``producers/bus_gps_producer.py``, ``producers/meter_producer.py``, ``producers/traffic_producer.py``, ``consumers/dlq_logger.py``, and ``consumers/enrichment_worker.py``.

Code Snippets

The producers now use ``acks=all``, ``enable.idempotence=True``, retries, and DLQ routing.

Screenshots

See ``screenshots/overview.svg``.

Results

The repository now includes the required producer, DLQ, and enrichment workflows.

Challenges

Aligning the Python producer semantics with Kafka's durability options locally.

Future Work

Connect the DLQ to replay and dead-letter recovery tooling.

Conclusion

Task B is implemented and documented.